

Cary Shu

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<https://csh20000.github.io/>

EDUCATION

University of Michigan

Ann Arbor, MI

M.S Electrical and Computer Engineering-Sig. Img. Proc. & Machine Learning Focus Aug 2024 - Apr 2025

GPA: 4.0/4.0

B.S.E Computer Engineering

Aug 2021 - Apr 2024

GPA: 3.941/4.0

Awards: University Honors, Dean's List, William J. Branstrom Freshman Prize, James B. Angell Scholar

EXPERIENCES

University of Michigan

Ann Arbor, MI

Research Assistant

Oct 2023 - Present

- Proposed and developed innovative Augmented Reality (AR) simulation tool and pixel difference IoU performance metric to remotely simulate and compare different SLAM algorithm traces.
- Collected and processed user trial data for audio-visual AI assisted task guidance on Hololens system.
- Developed rehab games in Unity with an LLM agent using OpenAI API to analyze patient performance.
- Utilized Azure and Firebase to host cloud anchors for multi-agent SLAM comparison.

Graduate Student Instructor for Digital Signal Processing

August 2024 - Dec 2024

- Teach Python, embedded systems, digital signal processing (DSP), and machine learning concepts.
- Exhibited excellent communication instructing over 30 students in the Major Design course for DSP.

Orafol Automotive Graphics

Lake Orion, MI

Software Engineer Intern

May 2022 - Aug 2023

- Developed Android application for the shop floor using C# Xamarin Forms and .NET with intuitive GUI using XAML, facilitating streamlined tasks for workers saving up to 45 minutes per shift.
- Implemented RESTful Web API using ASP.NET to connect to Microsoft SQL Server database, enabling real-time data retrieval and updates.

PROJECTS

Automatic Number Plate Recognition

Mar 2024 - Apr 2024

- Optimized image preprocessing and CNNs within Number Plate Recognition and Optical Character Recognition systems using OpenCV and PyTorch resulting in up to 88% accuracy. [\[code\]](#)

Advanced Driver Assistance Systems (ADAS) Simulation

Sept 2023 - Nov 2023

- Developed Embedded C code for Manual Control, Adaptive Cruise Control, and Auto-Steering on NXP S32 board, enabling precise control and real-time communication via CAN network in car simulations.

Virtual Piano

Sept 2023 - Nov 2023

- Tracked hand movements for a virtual piano system using Python with Numpy on Raspberry Pi.
- Implemented piano key recognition system using contour detection and thresholding with OpenCV.

Optical Flow Mapping and Prediction

Feb 2023 - Apr 2023

- Explored Lucas-Kanade, Gunnar-Farneback, and Horn-Schunck algorithms to build optical flow maps.
- Estimated and predicted motion by processing video data and using aggregate flow vectors in Matlab.

Room Occupancy Counter

Jan 2023 - Apr 2023

- Developed an efficient FSM-based counting algorithm using C, IR and ToF sensors, and an Arduino microcontroller with >95% accuracy for real-time data processing.

PUBLICATIONS

- Filippos Bellos, Yayuan Li, Ruey Day, **Cary Shu**, Jeffrey Mark Siskind, Jason J Corso, "Evaluating Interactive AI Agents for Human-in-the-Loop Physical Task Guidance", to be submitted
- Zijian Huang*, **Cary Shu***, Hang Qiu, Jiasi Chen, "ReplayAR: A Tool for Visual Evaluation of Mixed Reality", *ACM ImmerCom Workshop*, 2024. [\[pdf\]](#)[\[code\]](#)
- Tristan McCarty, Smrithi Surender, **Cary Shu**, Zijian Huang, Jiasi Chen, Elena Kokkoni, "Integrating Mixed Reality and Body Weight Support Technology for Immersive Pediatric Rehabilitation", *Conference of the IEEE Engineering in Medicine & Biology Society*, 2024. [\[pdf\]](#)

SKILLS

- Programming Languages: Python, C++, C#, C, Javascript, SQL, CSS, HTML, XAML
- Tools: OpenCV, Pytorch, ScikitLearn, .NET, WPF, React, Node.js, Swagger, Git, Arduino, FPGA, Soldering, Oscilloscope, Matlab, Unity, Windows, Linux (Ubuntu, Raspian)
- Languages: English (Native), German (Intermediate), Mandarin Chinese (Intermediate)